Name: ..... Class: ..... Date: .....

Lesson 3: Organelles

Worksheet:

Cells vs cities

In the city of Keystonia the main industry is the production of keystones. Everyone in the town is involved in the production of this one product. The blueprint design for keystones is kept in the council chambers. When needed, a photocopy of the design is made and then taken to the assembly factory to be made. The keystones are then transported along the roads to the post office where they are packaged into special boxes and sent out of the town. The energy for the factory is made by a set of solar panels. These panels store their energy in batteries until they are needed. To protect the city a large wall with a few guarded gates surrounds the town.

The manufacture of keystones in Keystonia is very similar to how a cell works. Most cells are involved in producing special proteins. The nucleus contains the DNA instructions to make the protein. Copies of the DNA are made and transported to the ribosome where the protein is made. The protein is transported through a series of tunnels called the endoplasmic reticulum. The Golgi apparatus/body package up the protein and send it out of the cell's membrane (the structure that surrounds and protects the cell).

Match up the following parts of the cell with their corresponding jobs in the production of keystones in Keystonia by drawing a line connecting the matching parts.

council chambers	mitochondria
assembly factory	chloroplasts
roads	endoplasmic reticulum
post office	nucleus
solar panels	golgi apparatus/body
batteries	ribosome
city wall	cell wall

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